

ZAHOORAHMED SAYYAD

+91 8767354046 | zahoor.adcet@gmail.com | linkedin.com/in/zahoorahmedsayyad | github.com/mrz004

EXPERIENCE

System Software Engineer

Astrome Technologies Pvt Ltd

Jan 2026 – Present

Bangalore, India

- Worked on products like GigaMesh, Modem and GigaSat
- Implemented the Software Update Manager for OTA updates across 4–5 Astrome products: a signed bundle of 5 packages is installed sequentially, and a failed package is reported to the user while its previous version stays intact and the others are unaffected
- Involved in the design of the GigaSat-2 satellite ground terminal
- Worked on integration of devices like LMX2572, GPS, LCD, matrix keypad, BNO086 IMU and other peripherals in the products

Competitive Programming Intern

HeyCoach Pvt Ltd

Dec 2024 – Mar 2025

Bangalore, India

- Assisted coaches as TA in live classes of 200+ learners
- Assisted in creating 10+ LeetCode-style problems
- Undertook 50+ one-on-one doubt-clearing sessions with learners

TECHNICAL SKILLS

Languages: C++ (11/14/17), C, Python, Bash, ARM Assembly

Embedded & Systems: RTOS (FreeRTOS), bare-metal, Linux kernel/drivers, interrupts, DMA, memory-mapped I/O, multithreading

Interfaces & Protocols: UART, SPI, I2C, CAN, USB, GPIO

Platforms: ARM Cortex-M, AArch64, STM32, ESP32, Raspberry Pi

Build & Debug: CMake, Make, GCC, GDB, OpenOCD, JTAG/SWD, spectrum analyzer, oscilloscope

Tools & Testing: Git, Docker, GoogleTest, CI (GitHub Actions)

PROJECTS

GigaSat | C++, Python, Linux, FreeRTOS, STM32

Apr 2026 – Present

Satellite ground terminal with flat panel patch antenna, auto satellite pointing and dual satellite support.

- Implemented a frame-transformation-based control feedback loop using BNO086 IMU and GNSS data at 20 Hz (matched to the slower motors) to lock antenna pointing on the satellite, designed for 1° pointing accuracy
- Integrated multiple devices like GPS, LMX2572, etc.
- Achieved beacon and DVB-S2 signal reception with 14 dB signal strength
- Calibrated and replaced the old IMU setup with a BNO086 IMU for better performance, rewriting the IMU controller

GigaMesh | C++, Aarch64, Zynq, Buildroot, Prime

Jan 2026 – Present

Dual E-band node supporting 1.2 Gbps throughput with 2x2 MIMO and 256QAM modulation.

- Implemented weighted round robin QoS and validated it with a network analyzer generating random packets, which were classified and dropped as expected
- Implemented BIST (POST) that checks voltage, temperature, clocks, LMX/LMK and other key parameters against expected ranges
- Implemented the software update manager and bandwidth calculation
- Fixed the micro tar to support the upgraded USTAR standard, allowing 2x longer names to be stored

Modem | *C++, Aarch64, Zynq, Buildroot, Prime*

Jan 2026 – Present

High throughput modem supporting 1.3 Gbps of data transmission.

- Integrated the matrix keypad and LCD display, including driver implementation
- Integrated GPS by implementing an NMEA parser from scratch

EDUCATION

Annasaheb Dange College of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering

Sangli, Maharashtra

Aug 2023 – May 2026

Latthe Education Society's Polytechnic

Diploma in Computer Engineering

Sangli, Maharashtra

Aug 2020 – Jun 2023

ACHIEVEMENTS

- GATE 2025 (Computer Science): qualified, score 356/1000
- IBM SkillsBuild Maharashtra State Level Hackathon 2025: Finalist
- LeetCode: 300+ problems solved, contest rating 1700+ (leetcode.com/u/mrz004)